IAP Proofs Workshop 2026: Real Analysis

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January 21, 2026

1 Introduction

Welcome to the second lecture of the IAP Proofs Workshop!

1.1 Logistics

The UMA proof workshop will consist of four lectures given over IAP. The schedule and content is as follows:

- Workshop 1: Introduction to Proofs - Wednesday 1/14
- **Workshop 2: Real Analysis - Wednesday 1/21**
- Workshop 3: Combinatorics - Wednesday 1/28
- Workshop 4: Graph Theory - Friday 1/30

Unless indicated at a later time, all lectures will be given from $5-6:30$ pm in room $4-131$. Please email umadei@mit.edu with any logistical questions. For questions about the contents of lecture 2, you can reach me at pengs14@mit.edu.

These materials are based on 18.s097, which ran over IAP 2021. Materials from this class can be found on the UMA website at https://uma.mit.edu/18.S097. This includes lecture notes, past handouts, and solutions to past problem sets.

1.2 Good Definitions

Today, I want to give you a taste of the field of real analysis, but more importantly, the heart of higher mathematics: good definitions. If you've taken a look at MIT's course 18 catalog, you'll notice that this is a prerequisite for many of the non-introductory 18 courses. That's because real analysis is where most math majors everywhere are introduced to *rigorous*

definitions (which leads to rigorous proofs). We certainly cannot talk about proving that no true Scotsmen put sugar in their porridge when we disagree on who counts as a true Scotsmen!

A disclaimer: writing down rigorous definitions will be the main focus of this lecture, so we won't be doing as much proving (ironic). A cautionary tale that I like to tell is that of the Italian school of algebraic geometry. The tl;dr is that their proofs weren't rigorous, and at first what they were proving was still true because their intuitions were mostly correct. However, as time went on, because it is rather difficult to intuit 4-dimensional space, they started proving things that were flat-out wrong. A big part of what wasn't rigorous was that they would say things (like 'generic point') without defining them. In the mid 1900's, many mathematicians worked hard to rigorize these definitions and build a grand working dictionary of algebraic geometry (see this for more details).

But that's enough about things that are not real analysis.

1.3 What is Real Analysis Exactly?

In short, it is the study of the real numbers, $\mathbb{R}$. For instance, when does a sequence (of real numbers) converge? What is the limit of a sequence? Perhaps another question that you should be asking is what are the real numbers, exactly?

As with much of mathematics, we study more than just the objects, but also functions between them. We find that this tells us a lot about the objects themselves. Thus, real analysis also (mostly) concerns itself with questions about such functions $f : \mathbb{R}^n \to \mathbb{R}^m$ (you can read this as 'f is a function from (as in, takes as input) n-tuples of real numbers to (as in, outputs) m-tuples of real numbers'). When are they continuous? differentiable? integrable?

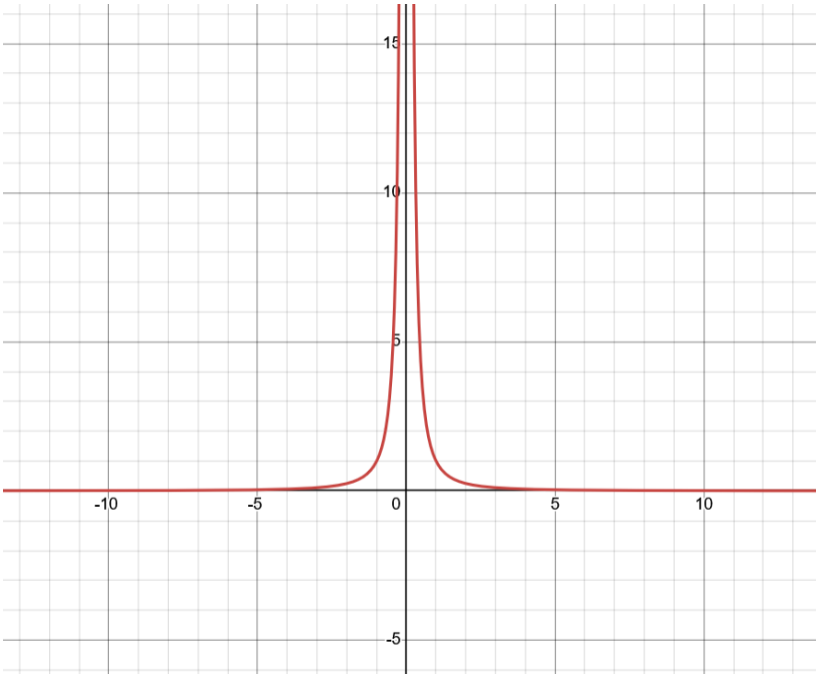
I use as primary reference [1] (which was the textbook for 18.100B when I took it) and [3] (OCW notes; more approachable and gives more intuition). I will not be introducing real analysis in the order a traditional course does. Instead, I would like to build one cohesive narrative surrounding limits and continuous functions. As a consequence we won't be talking much about the real numbers themselves unfortunately. Those interested may want to read the textbooks I've cited or take 18.100x for themselves for a treatment of the reals.

2 What is Continuity Exactly?

2.1 The calculus definition

Let's think back to when we first learned calculus. You may recall that a function is a continuous when we can draw its graph without lifting our pencil. You probably could do a good job at identifying when a function is continuous and when it is not. Let's try our hand at this function:

Example 2.1.1. Consider the function $f(x) = \frac{1}{x^2}$. Notably, it is not defined at $x = 0$.

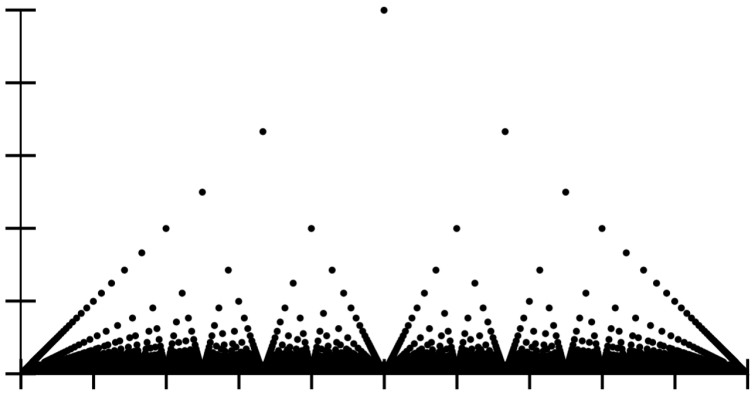


Would you say that f is continuous? Maybe not on $\mathbb{R}$ as it is not defined on all of $\mathbb{R}$ but what about on the set $\mathbb{R}\setminus\{0\}$? I certainly cannot draw it without picking up my pencil, but perhaps one could argue that on the intervals $(-\infty, 0)$ and $(0, \infty)$, I could draw the two parts without picking up my pencil, and conclude that this function is continuous on its set of definition.

We could extend the definition of continuity to subsets $S \subseteq \mathbb{R}$ by saying that they're continuous if we can draw its graph *on each connected component of* S without lifting our pencil. (The astute among you may notice that we haven't even defined what it means for a set to be connected.) Let's take a look at something even more pathological:

Example 2.1.2. Thomae's function, sometimes called the popcorn function, is defined as follows:

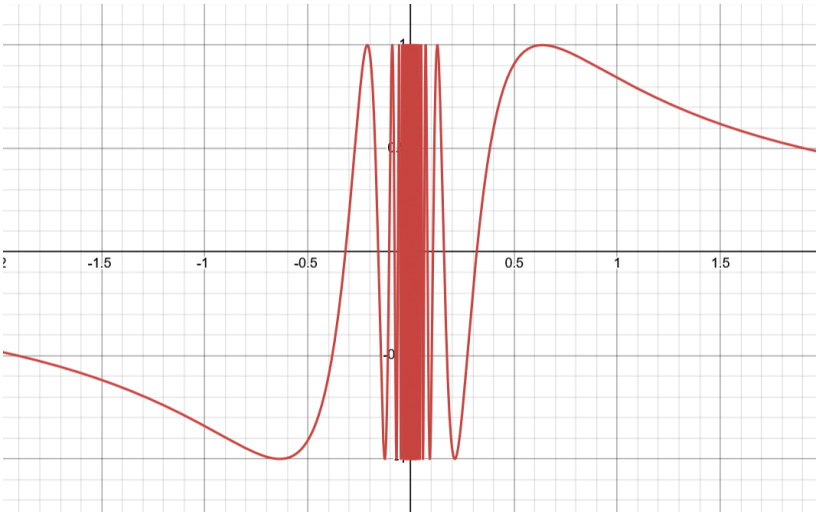
$$T(x) = \begin{cases} \frac{1}{q} & x = \frac{p}{q} \text{ rational} \\ 0 & x \text{ irrational} \end{cases}$$



I claim that this function is continuous on the irrationals but not on the rationals. Hopefully, this claim is not too outrageous. Certainly, it is the constant function on the irrationals, and we would expect constant functions to be continuous. On the other hand, there are tons of isolated points on the rationals. How can we use our definition previously to show these claims?

Example 2.1.3. The topologist's sine curve is defined as follows:

$$f(x) = \begin{cases} \sin(\frac{1}{x}) & x \neq 0 \\ 0 & x = 0 \end{cases}$$



Is this function continuous?

Hopefully I have given you enough reasons to doubt the definition of continuity given to you in calculus and motivate the reason behind our more rigorous definition. I will point out that this rigorization is nontrivial. Previous to the 19th century, mathematicians were treating continuity as an intuitive concept. The definition that I am about to present was first introduced by Karl Weierstrass (influenced by Bernard Bolzano's and Augustin-Louis Cauchy's formalizations) in the 1930's, a few hundred years after the theory of calculus was created (in the late 17th century).

2.2 Epsilon-Delta Continuity

A motivating story (I promise this is the last one), courtesy of [2]: Suppose that I am standing at the top of some building and that I am going to drop a ball. I have a very precise stopwatch and slow motion camera that is able to record the time it takes for the ball to drop to the ground up to any precision (but due to the finiteness of man-made instruments, we are unable to get the exact number). Let's call this quantity x. I would like to know how tall this building is. Let's call this quantity y. We know from physics that $y = gx^2/2$.

Let's say further that I want to measure y up to some precision. How precise do my measurements of x have to be in order to guarantee this? More fundamentally, how do I know that the small error I have in measuring x might not bork my calculation of y up? You might imagine another world where $y = T(x)$ where T is Thomae's function defined in Example 2.1.2. Suppose we measure $x = \frac{1}{2}$ when x actually is $\frac{1}{2} + \frac{\pi}{9999999999}$. Then, our calculated y would be $\frac{1}{2}$ when our actual value is 0. These are very different values! In this hypothetical world, we cannot trust that our measurements of x could predict y well

(ignoring the fact that any measurement of x we have is rational; the main point is that we cannot guarantee that if we measure x to some precision, we would know y up to some precision). The reason why we are able to perform the calculations in the first scenario is that $gx^2/2$ is a continuous function.

Definition 2.2.1. Let f be function from some subset of the real numbers $X \subseteq \mathbb{R}$ to $\mathbb{R}$. We say that f is **continuous at a point** $x_0 \in X$ if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in X$ satisfying that

$$|x - x_0| < \delta$$

we have that

$$|f(x) - f(x_0)| < \varepsilon$$

We say that f is **continuous on some set** $S \subseteq \mathbb{R}$ if it is continuous for all points $x_0 \in S$.

Remark 2.2.2. The ε here represents the precision in y we are looking for, and the δ represents the precision in x we need to have in order to achieve that.

Since this is a proofs workshop, let's warm up with a (not-so-trivial) proof that our favorite quadratic function is continuous:

Proposition 2.2.3. $f(x) = x^2$ is continuous on $\mathbb{R}$.

Proof. Fix some $x_0 \in \mathbb{R}$ and $\varepsilon > 0$. We would like to find some δ such that for all x such that $|x - x_0| < \delta$, we have

$$|x^2 - x_0^2| < \varepsilon$$

Let us work backwards to obtain this δ. We know that

$$\begin{aligned} |x^2 - x_0^2| &= |(x - x_0)(x + x_0)| \\ &= |x - x_0| \cdot |x + x_0| \\ &< \delta \cdot |x + x_0| \end{aligned}$$

We want the quantity on the right hand side to be less than ε. The annoying part is that x. If I turned it into a constant, I would be able to solve for δ. Let's do that! Let us enforce that $\delta < 1$. Then, (there is a little bit of casework with signs that you can work out yourself) $|x + x_0| < 2|x_0| + 1$ and we have

$$\delta \cdot |x + x_0| < \delta \cdot (2|x_0| + 1)$$

and so we notice that we can set $\delta = \min(1, \frac{\varepsilon}{2|x_0|+1})$ to satisfy the definition of continuity. □

Remark 2.2.4. We note that δ can (and usually does) depend on both the point x_0 and ε.

Let's move on to the claim in Example 2.1.2.

Proposition 2.2.5. The Thomae function is not continuous on the rationals and continuous on the irrationals.

Proof. First we note that this function is periodic with period 1. Thus, it suffices to consider points on the interval $[0, 1]$.

Let's start by showing that it is not continuous on the rationals. We note that the image of the rationals under T is the ordered set $\{\frac{1}{2}, \frac{1}{3}, \dots\}$. Thus, if we have some x_0 rational such that $f(x_0) = \frac{1}{q_0}$, we note that there are no $x = \frac{p}{q} \in \mathbb{R}$ such that

$$|f(x) - f(x_0)| = \left|\frac{1}{q} - \frac{1}{q_0}\right| < \frac{1}{q_0} - \frac{1}{q_0 + 1}$$

so we can take $\epsilon = \frac{1}{q_0} - \frac{1}{q_0+1}$.

Now let us show that the Thomae function is continuous on the irrationals. Fix x_0 some irrational and $\varepsilon > 0$. Intuitively, we want to pick a δ so small that none of the rational functions with small denominators are within a δ-distance from x_0. Precisely, we can find (via the Archimedean property of the real numbers, but for now I'm sure you can believe me on this one) some q such that

$$\frac{1}{q} < \varepsilon$$

Then, we can pick some δ such that

$$\delta = \min\left\{\left|x_0 - \frac{a}{b}\right| : 1 \le b \le q, 0 \le a \le b\right\}$$

We note that this set is finite and does not contain 0 as x_0 is not rational. Thus, $\delta > 0$. By construction, for any point $\frac{a}{b}$ such that $|x_0 - \frac{a}{b}| < \delta$ satisfies that $b \ge q$ so $|f(\frac{a}{b}) - f(x_0)| = \frac{1}{b} \le \frac{1}{q} < \epsilon$ as desired. □

Remark 2.2.6. When doing epsilon-delta proofs, there is nothing stopping you from picking δ as small as you want. You are by no means obliged to use the 'largest δ that still works' as I may be doing in the above proof.

3 Metric Spaces

3.1 Multivariable Continuity

We've defined continuity for single variable functions. What about multivariate functions $f : \mathbb{R}^n \to \mathbb{R}^m$? Let's return to our ball-drop experiment.

Let's further suppose that instead of just calculating how tall this building is, I want to calculate exactly how much ball traveled as a three-dimensional vector. Additionally, I'll also measure wind speed and direction to calculate this. We'll assume again that I can measure these quantities up to arbitrary precision, and I want to know the resulting displacement of the ball up to some margin of error. How would I encode this margin? One way is to say that the calculated value is within some Euclidean distance away from the actual.

So to generalize our notion of continuity, we'll replace the 1-dimensional distance with the n-dimensional Euclidean distance, which we abbreviate as d_n:

$$d_n(\vec{x}, \vec{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

In the future, we'll drop the vector notation and just use x, y to mean arbitrary points in $\mathbb{R}^n$.

Definition 3.1.1. Let $f : \mathbb{R}^n \to \mathbb{R}^m$. We say that f is **continuous at a point** $x_0 \in \mathbb{R}^n$ if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in \mathbb{R}^n$ satisfying

$$d_n(x - x_0) < \delta$$

we have that

$$d_m(f(x), f(x_0)) < \varepsilon$$

We say that f is **continuous on some set** $S \subseteq \mathbb{R}^n$ if it is continuous at all points $x_0 \in S$.

Remark 3.1.2. Note that we are using two different distances in two different spaces here! I added the subscript to emphasize this point. Some authors like to write d_X to denote the metric on the space X.

In mathematics, generalization is a very powerful tool for us to strip away extraneous constraints and apply the same general proof to many situations. You might notice that all we need to use this epsilon-delta definition of continuity is some notion of distance. If we use some other metric, we can define continuous functions similarly! For example, we may use a different unit of measurement. We might even want to talk about continuous functions

on manifolds such as a sphere. Here, the notion of distance is a little different. We haven't proven anything about continuous functions yet, but you will in the problem set (and you definitely will in a real analysis class), and these generalizations allow you to apply those properties to other metric spaces.

3.2 Generalizing Distance

Definition 3.2.1. A **metric space** is a set X along with a function $d : X \times X \to \mathbb{R}$ (called the **metric**) satisfying the following properties for all $x, y \in X$:

- (Nonnegativity) $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$,
- (Symmetry) $d(x, y) = d(y, x)$, and
- (Triangle Inequality) $d(x, y) \leq d(x, z) + d(z, y)$ for all $z \in X$.

You can check for yourself that the Euclidean distance on $\mathbb{R}^n$ is a metric. Here are some examples that you might not have seen before. As we go through these, prove to yourself that these are metrics.

Example 3.2.2. The complex norm gives a metric on $\mathbb{C}$. Suppose we have some number $z = x + iy$ where $x, y \in \mathbb{R}$. Then, we define

$$|z| = \sqrt{x^2 + y^2}$$

We define the distance between two points to be the norm of their difference:

$$d(z_1, z_2) = |z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Example 3.2.3. We can endow any set with the discrete metric, which is a silly metric defined as follows:

$$d(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

Example 3.2.4. Let $C([0, 1], \mathbb{R})$ be the set of continuous functions $f : [0, 1] \to \mathbb{R}$. We define a metric as follows:

$$d(f, g) = \sup_{x \in [0,1]} |f(x) - g(x)|$$

You can take sup to mean max but for infinite sets. More precisely, it is the least element (not necessarily in the set) that is larger than all elements of the set.

This metric is actually used to prove Picard-Lindelöf, which is a statement of the existence of an unique solution to a ODE under some conditions. In the proof, you apply Banach fixed point on a contraction that is continuous in the sup norm.

Example 3.2.5. As a more number-theoretical example, we have the p-adic norm on the rational numbers $\mathbb{Q}$. Fix some prime p and integer a. The p-adic valuation $v_p(a)$ is the largest power of p that divides a. We define the norm of a rational number to ne

$$\left|\frac{a}{b}\right|_p = p^{-(v_p(a)-v_p(b))}$$

Then, we obtain a metric with the norm

$$d(x,y) = |x-y|_p$$

Now that we've looked at some metrics, we can copy and paste our definition of continuity and apply it to all metric spaces.

Definition 3.2.6. Let X, Y be metric spaces and $f : X \to Y$. We say that f is **continuous at a point** $x_0 \in X$ if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in X$ satisfying

$$d_X(x, x_0) < \delta$$

we have that

$$d_Y(f(x), f(x_0)) < \varepsilon$$

We say that f is **continuous on some set** $S \subseteq X$ if it is continuous at all points $x_0 \in S$.

Note that different functions are continuous under different metrics!

Example 3.2.7. Let X be a metric space equipped with the discrete metric and Y an arbitrary metric space. Then, any function $f : X \to Y$ is continuous: we can just pick $\delta = 1$ and the only point with distance less than 1 from a point is that point itself.

In the next couple sections, we'll be talking about preimages. I'm going to leave the definition here before we get into that.

Definition 3.2.8. Let $f : X \to Y$ be a function on two sets X, Y. Let $S \subseteq Y$ be a subset. The **preimage of** S **under** f is the set

$$f^{-1}(S) := \{x \in X : f(x) \in S\}$$

3.3 Open Balls and Open Sets

You might notice that we can reformulate the definition of continuity by drawing spheres centered around the point in question. However, depending on what metric we use, the set of distance ε away might not actually look like the usual notion of a sphere. Instead, we call them balls.

Definition 3.3.1. Let (X, d) be a metric space. We define the **open ball with radius** $r \in \mathbb{R}$ **around the point** $a \in X$ to be the set

$$B_r(a) := \{x \in X : d(x, a) < r\}$$

Remark 3.3.2. We can define a closed ball if we use $\leq$ instead of $<$, but we won't use those. In the future, when we say ball, we shall mean open ball.

I like to visualize these open balls (and open sets in general, which we'll see below) to be fuzzy: there is no hard boundary for where it stops, and if you're within the ball, you can always inch slightly a tiny bit farther away from the center and you'd still be in the ball.

You might wonder (after trying to substitute $B_\delta(x_0)$ and $B_\epsilon(f(x))$) if the preimage of an open ball under a continuous map f is an open ball. The answer is not quite. What is true is that for some ball $B_\epsilon(y) \subseteq Y$, its preimage $f^{-1}(B_\epsilon(y))$ *contains* an open ball around x – convince yourself that this is true from the definitions.

To demand that the preimage of open balls is an open ball is too strict of a condition. Instead, let's relax our constraints to unions of open balls, which we call open sets.

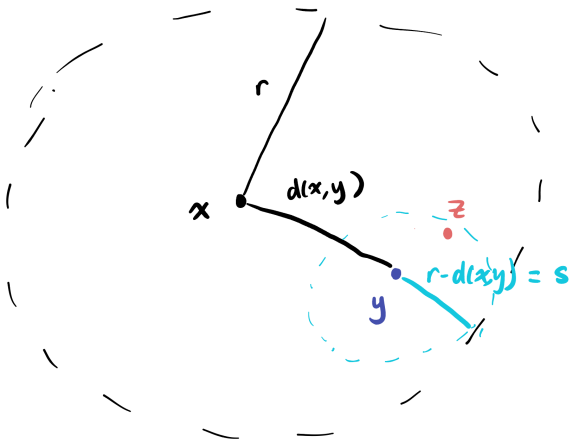
Definition 3.3.3. A subset $U \subseteq X$ is **open** if for every point $p \in U$, there exists some ball B such that $x \in B \subseteq U$.

Let's do a quick sanity check to get used to working with this definition.

Proposition 3.3.4. Open balls are open.

Proof. Suppose we have some ball $B_r(x)$ and some point $y \in B_r(x)$. Then, let $s = r - d(x, y)$. We claim that

$$y \in B_s(y) \subseteq B_r(x)$$



This amounts to showing that all points in $B_s(y)$ also lie in $B_r(x)$. Let $z \in B_s(y)$, so $d(z, y) < s$. Then, by triangle inequality,

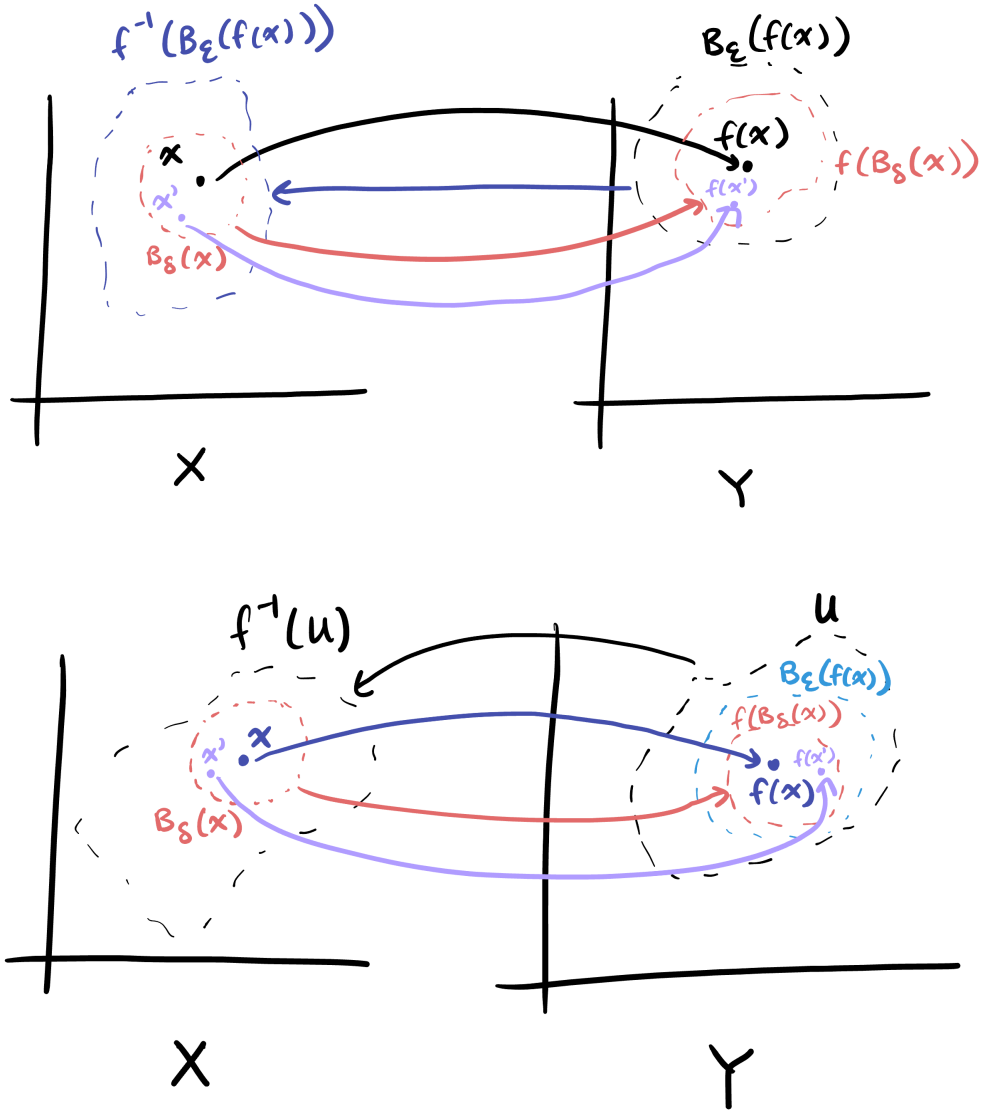
$$\begin{aligned} d(x, z) &\leq d(x, y) + d(y, z) \\ &< d(x, y) + r - d(x, y) \\ &< r \end{aligned}$$

so $z \in B_r(x)$ as desired. □

Now let us prove what I claimed earlier, that the preimage of an open set under a continuous function is an open set. In fact, the converse is true as well!

Theorem 3.3.5. A function $f : X \to Y$ is continuous if and only if for every open set $U \subseteq Y$, $f^{-1}(U)$ is an open set.

Proof. ($\implies$) Suppose that f is continuous and let U be some open set. Let $x \in f^{-1}(U)$. We want to show that there is an open ball around x that is contained in $f^{-1}(U)$. Pick some $\varepsilon > 0$ such that $B_\epsilon(f(x)) \subseteq U$. Then, since f is continuous, there exists a $\delta > 0$ such that for all $x' \in B_\delta(x)$, $f(x') \in B_\varepsilon(f(x)) \subseteq U$. In other words, $x' \in f^{-1}(U)$ for all such x', and so $B_\delta(x) \subseteq f^{-1}(U)$.



($\Longleftarrow$) Suppose that $f^{-1}(U)$ is open for all open sets $U \subseteq Y$. Fix some $\varepsilon > 0$ and $x \in X$. Consider the ball $B_\varepsilon(f(x)) \subseteq Y$ which is open. Its preimage is also open. Thus, we may find some open ball around x, $B_\delta(x)$, that is contained within $f^{-1}(B_\varepsilon(f(x)))$. We then note that for all x' such that $d(x', x) < \delta$ (ie. $x' \in B_\delta(x)$), we have that $f(x') \in B_\varepsilon(f(x))$, or $d(f(x'), f(x)) < \varepsilon$.

□

References

[1] Walter Rudin. *Principles of Mathematical Analysis*. McGraw-Hill, Inc., 1976.

[2] Timothy Gowers. *One way to arrive at the definition of continuous functions*. URL: https://www.dpmms.cam.ac.uk/~wtg10/continuity.html.

[3] Jirí Lebl. *Basic Analysis I*. URL: https://ocw.mit.edu/courses/18-100a-real-analysis-fall-2020/mit18_100af20_basic_analysis.pdf.